Titanic Dataset Analysis Exercise

Question 1: Using the Titanic dataset from Seaborn, create a stacked bar chart that shows the distribution of passenger class by sex.

Instructions:

i. Load the titanic dataset from seaborn.

ii. Handle the missing values accordingly.

- Identify which columns have missing values
- Document the percentage of missing data
- Apply appropriate strategies for handling missing values
- Justify your approach

iii. Create a stacked bar chart showing the count of male vs female passengers for each class (First, Second, and Third).

- Group the data by passenger class and sex
- Create a stacked bar visualization
- Ensure the chart clearly shows the distribution

iv. Make sure your chart has:

- A clear title
- Labelled axes
- A legend showing the sex categories
- Professional appearance with appropriate colors

v. Create a violin plot showing age distribution for each passenger class (First, Second, and Third).

- Handle missing age values appropriately
- Show the distribution shape for each class
- Include statistical indicators (mean/median)

Question 2: Fare Analysis Exercise

a. Write Python code that creates a new feature (variable) called 'fare_age_ratio' calculated as fare divided by (age + 1). Adding 1 to age prevents division by zero for infant passengers.

b. Group the data by passenger class and embarkation port (embark_town), and calculate:

- Average fare per person
- Average

- Total number of passengers who survived

Question 3: Advanced Visualization and Analysis

Create a comprehensive dashboard with the following components:

a. A grouped bar chart comparing survival rates across:

- Passenger classes
- Sex
- Age groups (create categories: Child (0-12), Teen (13-19), Adult (20-59), Senior (60+))

b. A scatter plot showing the relationship between:

- Age (x-axis)
- Fare (y-axis)
- Color-coded by survival status
- Size representing the passenger class